

RapidSmith: A Rapid Prototyping Tool for FPGA Designs

A research-based, open source electronic design automation (EDA) tool written in Java for modern Xilinx FPGAs

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The design and verification of field programmable gate arrays (FPGAs) can be very tiring work due to the time-consuming nature of FPGA compilation processes such as synthesis, mapping, placement and routing. This severely affects the productivity of an FPGA design engineer.

The BYU RapidSmith Project is a set of tools and application programming interfaces (APIs) written in Java language, with the goal of providing users with an easy-to-use platform to work on modern Xilinx FPGAs. Or in simple words, RapidSmith is a do-it-yourself CAD tool for the FPGA models of Xilinx. It offers a compact and fast device database with several APIs to enable different functions for Xilinx devices. RapidSmith project was developed by Brigham Young University, USA. It offers an excellent platform for academia to try out experimental ideas and algorithms for all fields of FPGA CAD research.

RapidSmith is based on Xilinx design language (XDL), which is a fully featured physical design language that provides human-recognisable format of Xilinx's proprietary native circuit description (NCD).

RapidSmith eases the difficulties of using XDL. It is mainly aimed at

TABLE I System Requirements	
Operating system	Windows 7, Windows Vista, Windows XP Linux Mac OS X (RapidSmith works, but Xilinx tools do not run on OS X)
Disk space	400MB
Software	Xilinx ISE Design Suite 11.1 or higher ISE 10.1.03 or earlier for legacy (outdated) devices Java development kit (JDK) 1.6 or higher
Supporting JAR	Included: Caucho Hessian Implementation JAR v.4.0.6 Included: Qt Jambi, the Qt library for Java Included: JOpt Simple Optional: JavaCC (if the user wants to change the XDL design parser. There is also a plug-in for Eclipse for JavaCC which helps compile .jj files)

the use by academia in all fields of FPGA CAD research, and lets the researchers import XDL/NCD files, manipulate, place, route and export designs and perform a variety of design transformations. It also contains packages which can parse or export bitstreams at the packet level, and represent the frames and configuration blocks in the provided data structures. It can resolve, manipulate and export bitstreams according to documented methods of Xilinx.

RapidSmith is not a replacement for Xilinx ISE design suite. It can be used only with a current valid licence to Xilinx tools installation. Any person with a basic knowledge in programming and Java and a good understand-

ing of Xilinx FPGAs and XDL can easily operate this tool.

Licence

BYU RapidSmith Tools is a free software which comes under general public licence (GNU). One can redistribute and/or modify it under the terms of GNU GPL. The main purpose of this package is that it can be used as a research

tool, and should not be used for designs bound for commercial purpose.

Supported devices

RapidSmith is supported in Virtex 4 and 5 families in all forms and applications. It can create device files for all modern Xilinx FPGA families and hence can support, though to a lesser extent than Virtex 4 and 5, for other Xilinx FPGA families too. In the case of legacy device families, the design suite version ISE 10.1.03 or earlier may be required. The compatible features of various families are shown in Table II.

Installation

First you need to ensure that the Xilinx tools and Java development